1.5 Digital text and numbers

In this lesson

You will learn:

- how numbers are stored as digital data
- how text is stored as digital data.

What is a value?

You use numbers every day. Some of those numbers are used in calculations. For example, you add or multiply the numbers together. This type of number is called a **value**.

Other numbers are not values. A phone number is not a value. You don't add phone numbers together or subtract one phone number from another.

When these two types of number are stored as digital data they are stored in different ways.

Activity

Discuss with your group which of these numbers are values and which are not values. Report your findings to the class.

a your house number

b the number of marks you get in a test

c your age

d the number of points your team scores in a basketball game.

Other types of content

Number values are stored inside the computer using the binary number system. But what about other types of content? For example:

- text
- pictures
- sounds.

All of these types of content must be turned into numbers. Then the computer can store these numbers using the binary number system.

In the rest of this unit you will learn how the computer stores text, images and sounds as number values.

Storing text using ASCII code

Text characters include letters of the alphabet, punctuation marks, and other characters that you can type on a standard keyboard, such as spaces and maths symbols.

The computer uses a number code to represent these text characters. There is a common code used by almost all computers. It is called ASCII (ask-ee). Here are number codes for the letters a to g in lower case. Different codes are used for upper-case letters.

Character	Number code
a	97
b	98
c	99
d	100
e	101
f	102
g	103

How the computer uses ASCII code

When you press a key on the keyboard it sends a signal to the processor. The signal varies according to which character you selected on the keyboard.

When the processor receives the signal it stores the character using ASCII code. It stores the binary number. So, for example, the letter 'a' is stored as the binary number 01100001. The letter 'b' is stored as binary number 01100010, and so on.

Each ASCII code takes up exactly 1 byte (8 bits).

Activity

Make an ASCII table.

a Copy the ASCII table in your book.

b Extend the table to show the whole alphabet from a to z.

c Add an extra column to the table to show the binary number for each character.

Use the ASCII table.

- Write your first name in ASCII using decimal values.
- Write your first name in ASCII using binary values.

Storing other characters

You have used ASCII to represent lower-case letters from a to z. ASCII can also be used to store upper-case letters and other keyboard symbols such as punctuation marks.

The next table shows the ASCII codes for some common keyboard characters.

Keyboard character	ASCII code (decimal)	ASCII code (binary)
Space	32	
Comma	44	
Full stop	46	

Activity

a Copy and complete the ASCII table to show the binary codes for the three keyboard characters.

b Here is a message in decimal ASCII. What does it say?

101 118 101 114 121 032 099 111 109 112 117 116 101 114 032 117 115 101 115 032 116 104 105 115 032 099 111 100 101

c Here is a message in binary ASCII. What does it say?

01111001 01101111 01110101 00100000 01110111 01101001 01101110

Numbers in ASCII

The digits from 0 to 9 also have ASCII codes. The ASCII code for a digit is not the same as its number value.

Keyboard character	ASCII code (decimal)	ASCII code (binary)
0	48	
1	49	
2	50	
3	51	
4		
5		
6		
7		
8		
9		

The computer can store a number using ASCII or as a number value. Different types of software store numbers in different ways.

- If you type the number 39 into a word-processed document, the computer will store the ASCII codes for 3 and 9. You cannot do a calculation using a word processor.
- If you type the number 39 into a spreadsheet, the computer will store the number value 39. You can do calculations using a spreadsheet.

Activity

Copy and complete the number table to show the ASCII codes for all the digits from 0 to 9.

A problem with ASCII

There are only 256 characters in ASCII. ASCII was originally created to convert from English into binary. However, people around the world need to use computers in their own language.

An improved code called **Unicode** was invented in 1991. Around 110,000 characters are available in Unicode, including the ASCII codes. Unicode has codes for characters in Arabic, Mandarin and Japanese, as well as many other languages.

- ASCII uses a single byte to store the code for each character. The maximum number of characters a single byte can hold is 256.
- Unicode uses more than one byte to store characters. Two bytes joined together can hold 65,000 characters. Three bytes joined can hold nearly 17 million characters.

Activity

Work with a partner. Write a short message in ASCII with about 10 characters and no punctuation. Give your message to your partner to decode. Work together to check you have coded and decoded each message correctly.

Extra challenge

Search the web to find a full ASCII character code table. Make sure that the table includes the binary codes. Look at the ASCII codes for lower-case and upper-case letters. Explain how they are different.

Test

1 What does 24 look like when it is stored as:
   a a value in binary?
   b ASCII?
2 When you type a character on your keyboard, what data is sent to the computer?
3 Why is ASCII limited to 256 characters?
4 What advantages are there to using Unicode instead of ASCII?



1.6 Sound and vision

In this lesson

You will learn:

- how images are converted into digital data
- how audio sounds are converted into digital data.

Everything you store on a computer must be stored as digital data. In the last lesson you learned how numbers and text are stored as digital data. Images, sound and video are also converted into binary so that a computer can store and use them.

Digital images

When you look at a photo on your computer screen it looks like the real world. In fact, it is an image made up of tiny squares called **pixels**. The word pixel is short for 'picture element'.

The pixels are organised in a grid of rows and columns. The picture is like a spreadsheet, but the cells contain colours not numbers. Each pixel stores a single colour.

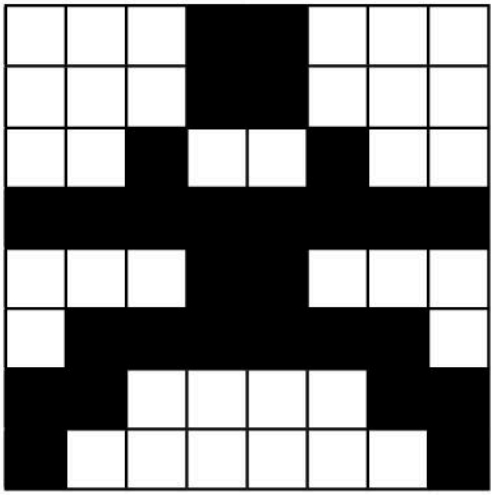
Pixels are so small that you cannot see the individual squares. Your brain blends the individual colours together to create a realistic image. An image created in this way is called a bitmap. Most images you see on computer screens are bitmaps.



Each colour the computer uses has its own binary code. The computer uses a colour code chart to convert the colour of each pixel into a binary code.

Example 1: storing two colours

A simple image has been created on a square grid measuring 8 × 8 pixels. The image uses only two colours – black and white. All the information the computer needs to store the information about each pixel can be stored in a single bit.



1	1	1	0	0	1	1	1
1	1	1	0	0	1	1	1
1	1	0	1	1	0	1	1
0	0	0	0	0	0	0	0
1	1	1	0	0	1	1	1
1	0	0	0	0	0	0	1
0	0	1	1	1	1	0	0
0	1	1	1	1	1	1	0

If the pixel is white, the computer stores a 1 in the bit. If the pixel is black, the computer stores a 0. The computer can store all the information it needs about each line in this simple image using a single byte. The information looks like this:

11100111 11100111 11011011 00000000 11100111 10000001 00111100 01111110

Activity

Draw a blank 8 × 8 pixel grid. Recreate the image stored in the digital data below. Work from the top left of the grid to the bottom right. 1 is a white square and 0 is black.

11111111 11100011 11011101 10101010 10111110 10100010 11011101 11100011

Images with more colours

Most images use more than two colours. In Example 1, a single digit was used to store information about the colour of a pixel. A single digit can store just two values. To store more colours, a computer uses more bits.

Example 2: storing more colours

In this example, the computer uses two bits to store the colour information about a single pixel. Using two bits means four colour codes can be used: 00, 01, 10 and 11. The codes used to store the colours in this example are: 00 black, 01 red, 10 blue and 11 white. The method is the same as in Example 1. Using more bits means you can use more colours in an image.

11	11	00	00	00	11	11	11	00	00	00	11	11
11	00	01	01	01	00	11	00	01	01	01	00	11
00	01	01	01	01	01	00	01	11	11	01	01	00
00	01	01	01	01	01	01	11	10	01	01	01	00
00	01	01	01	01	01	01	11	01	01	01	01	00
00	01	01	01	01	01	01	01	01	01	01	01	00
11	00	01	01	01	01	01	01	01	01	01	00	11
11	11	00	01	01	01	01	01	01	01	00	11	11
11	11	11	00	01	01	01	01	01	00	11	11	11
11	11	11	11	00	01	01	01	00	11	11	11	11
11	11	11	11	11	00	01	00	11	11	11	11	11
11	11	11	11	11	11	00	11	11	11	11	11	11
11	11	11	11	11	11	11	11	11	11	11	11	11

True colour

The examples in this lesson are simple images to explain how images can be stored as digital data. For most images, a computer uses more than two bits to store colour information.

For simple images like icons and emojis, the computer uses a byte (8 bits) to store colour information. A byte can store 256 different colours.

Photos need more than 256 colours to look realistic. A computer uses a method called **true colour** to store digital data about photos. True colour uses three bytes to store the information about a single pixel. True colour allows nearly 17 million colours to be used.

Realistic images

Using more colours makes an image look realistic. This is called **colour depth**. Adding more colours increases the colour depth.

Another way to make images more realistic is to use more pixels. This is called **resolution**. Using more pixels to display an image gives high-resolution images.



Low resolution



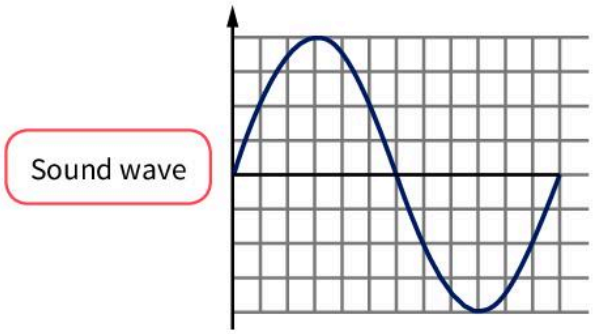
High resolution

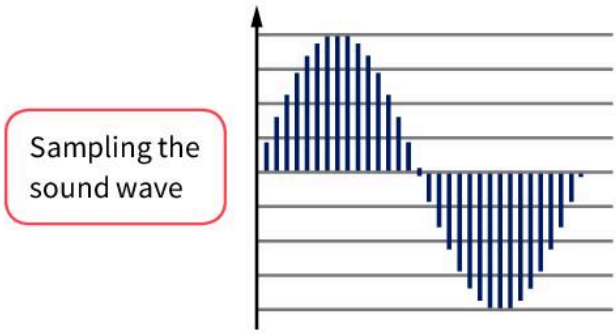
Digital sound

If you press a piano key, a hammer strikes a string and the string vibrates. The vibration is a wave of sound that passes through the air to our ears. The sound wave is continuous and smooth.



A computer cannot save continuous data. The computer must break up continuous data into chunks that can be stored in bytes in its memory. This process is called **sampling**. A sample is a slice of the sound taken at a moment in time.





When sampling, the computer takes measurements of the continuous sound wave at regular intervals throughout the recording. The computer stores the measurements as digital data. The number of samples recorded per second is called the sampling rate. A piece of digital music is sampled around 44,000 times per second.

The sampling process never captures sound exactly. There are gaps between samples where sound is missed, but the gaps are so small that the digital sound appears continuous when we listen to it.

High-quality audio recordings have a high sampling rate. More samples are taken per second during the recording.

Digital video

Digital video is created in the same way as audio. Sampling is used to capture slices of an event that happens continuously. The slices are called **frames**. When frames are displayed quickly one after the other, we see an image moving as it would in real life.

Earlier in this unit you created a flick book animation. The activity simulated the way that video is captured and displayed as digital data.

When a video is stored as a digital file, the images and audio are saved separately using the methods described in this lesson.

Activity

Create your own two-colour picture in an 8 × 8 grid.

Convert each line in your grid into a byte: white = 1, black = 0.

Exchange your binary code with a partner.

Use the binary code to redraw your partner's image.

Extra challenge

Create an information sheet that explains how sampling is used to capture and store audio in a digital data file. Search the web to find an image and more information about sampling.

Test

1. Explain in your own words how pixels are used to make a digital image.
2. Describe how sampling is used to capture audio on a computer.
3. How are colour depth and resolution used to create high-quality images?
4. High-quality image files are larger than low-quality files. Why is that?